ORIGINAL ARTICLE

Systems engineering; MBSE; nanosatellites; electrical power systems

Model-Based Systems Engineering for Nanosatellite Electrical Power Systems: Generality, Value Claims, and Pre-Physical Verification

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Purpose: To distinguish nanosatellite MBSE evidence from EPS-relevant evidence and model-linked V&V before physical development.

Methods: A protocol-specified review searched Scopus and IEEE Xplore; the final search/reassessment record was closed through 24 April 2026. Forty extracted reports (21 core and 19 supporting) yielded 270 nested claim records.

Results: Two reports and 17 of 270 claims were EPS-specific. Direct pre-physical mechanisms were coded in 26 reports, V&V artifacts in 24, and integration/test readiness in 10; these fields are distinct from a derived mixed-source linkage synthesis.

Conclusion: The corpus reports reusable system-level models and pre-physical analysis, but provides limited reported evidence for direct EPS outcomes, baseline-backed value, and model-to-test closure.

KEYWORDS
nanosatellites; CubeSats; electrical power systems; model-based systems engineering; verification and validation; systematic literature review

Abbreviations: EPS, electrical power system; IVV, integration, verification, and validation; MBSE, model-based systems engineering; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; RQ, research question; SIL, software-in-the-loop; HIL, hardware-in-the-loop; V&V, verification and validation

1 | INTRODUCTION

Broader reviews of space MBSE provide useful context on implementation and gap framing. In particular, Buruso et al. (2026) offer a methodological and positioning reference, but their broader synthesis is not evidence about the individual nanosatellite reports assessed here. An EPS-relevant nanosatellite evidence chain is needed because system-level MBSE models, mission-level outcome claims, and direct power-subsystem evidence cannot be assumed equivalent.

This review contributes a protocol-specified, controlled synthesis that distinguishes report-level context from nested claims, EPS-relevance classifications from EPS-specific claims, and direct V&V source fields from derived linkages. Its protocol-specified research questions were:

RQ1. How can nanosatellite missions and EPS be modeled in a general manner with minimal knowledge about their behavior?

RQ2. What are the claimed outcomes related to reliability, reproducibility, and integrability derived from EPS modeling in nanosatellite missions?

RQ3. How have verification procedures been integrated into EPS models prior to physical development?

2 | METHODOLOGY

2.1 | Protocol, eligibility, and research questions

This review followed a predefined protocol for nanosatellite-class missions, MBSE interventions, eligibility decisions, extraction, quality appraisal, and synthesis. Eligibility was limited to English conference and journal papers published from 2013–2026 that were open access or available through the TEC libraries. The population comprised CubeSat and nanosatellite-class missions, excluding larger spacecraft and smaller educational platforms such as CanSats. RQ1 considered system- or EPS-level MBSE rather than other subsystem-level or non-MBSE models. For RQ2, outcomes had to be attributed to MBSE artifacts, model-enabled activities, or model-enabled process changes rather than to an EPS technology choice. For RQ3, direct evidence required reported integration of IVV or verification procedures with an MBSE model before physical development.

2.2 | Information sources, selection, and appraisal

Candidate searching used Scopus and IEEE Xplore, and the final search/reassessment record was closed through 24 April 2026. Four concept clusters combined (i) general nanosatellite/CubeSat MBSE reference models, templates, behavior, and mission architecture; (ii) EPS, power, and electrical architecture; (iii) verification, validation, IVV, and test; and (iv) outcomes and integration, including traceability, reproducibility, data formalization, and MBSE/PLM linkage. Complete source-specific strings, filters, dates, yields, recovery routes, and stopping rationale are provided in the Supporting Information.

Exact-title Semantic Scholar metadata/abstract enrichment and targeted Google Scholar fallback and citation chasing informed assessment or recovery of specific records; they were not broad additional database searches. A total of 199 records were merged and deduplicated to 81 unique candidates. Screening applied the protocol boundaries and recorded rationales. Core reports directly informed the RQ synthesis, whereas supporting reports provided relevant context on toolchains, lifecycle continuity, requirements, project practice, or V&V.

Quality appraisal used eight protocol checklist items, scored No = 0, Insufficiently = 0.5, and Yes = 1, covering

venue credibility, limitations, context, MBSE detail, outcome linkage, outcome evidence, traceability, and reproducibility or toolchain reporting. Scores were descriptive and contextual: no quantitative sensitivity analysis was performed, and no score excluded a report.

2.3 | Extraction and synthesis safeguards

A controlled codebook captured RQ relevance, EPS relevance, model maturity, lifecycle stage, model artifacts, reuse mechanisms, approach classification, interoperability, V&V artifacts and IVV level, outcomes and evidence, governance and value realization, and the secondary gap mapping. Screening and extraction decisions used recorded rationales, source quotations, source-report cross-checks, and the controlled codebook. Only controlled codes contributed to aggregate counts; qualitative observations informed interpretation without creating new memberships.

A focused codebook-refinement review of generality/reuse, IVV level, and reliability subtype used isolated model-assisted per-report passes. The allowed labels and field coverage were validated, but this process was not an independent second assessment and no agreement statistic was calculated.

The analytic units were **40 included reports**, not 40 independent implementations. The **270 claim records are nested within reports**; accordingly, neither report totals nor claim totals estimate independent deployments. Study multi-select counts are **report memberships**, whereas claim multi-select counts are **claim memberships**. Model maturity is multi-label and is reported as co-occurring labels, not as an ordinal highest-maturity scale.

All status distinctions were retained. Field-status *not reported* denotes source silence, whereas field-status *none* denotes explicit absence or no use. A selected normalized *not reported* category is distinct from field-status *not reported*, and an unassigned value has no selected controlled label. Five baseline-comparator records contained free-text values outside the controlled vocabulary and were retained as *unmapped/invalid controlled value*; they are not author-reported statuses.

The synthesis is descriptive. Cross-tabs show controlled-code co-reporting only, not causal linkage. In particular, lifecycle stage $\times$ IVV level does not establish longitudinal progression or a direct linkage, and EPS-element $\times$ V&V-artifact co-reporting does not demonstrate element-specific verification. Narrative synthesis therefore interprets patterns conservatively, with quality and extraction-confidence ratings used to contextualize the evidence. Supporting Information includes the protocol, complete search record, controlled codebook, included-report inventory, aggregate tables, and claim-level evidence records.

3 | RESULTS

3.1 | Search, report characteristics, quality, and extraction confidence

The 199 records were merged and deduplicated to 81 candidates; 37 were excluded at screening and 44 were sought for retrieval. The retrieval candidates comprised 22 core and 22 supporting reports. One core and three supporting reports were not retrieved, leaving 21 core and 19 supporting extracted reports (40 in total). The included units are reports, not independent implementations. Figure 1 presents the selection flow.

The corpus contains 21 core and 19 supporting reports, published from 2013 to 2025. Quality scores ranged from 5.0 to 7.5 (mean 6.21): 3 reports scored 5.0, 6 scored 5.5, 13 scored 6.0, 9 scored 6.5, 7 scored 7.0, and 2 scored 7.5. Extraction confidence was high for 29 reports and medium for 11. RQ relevance was concentrated in RQ1; RQ2 was usually partial because outcome claims were more often reported than directly evaluated, and RQ3 was frequently partial because model-linked pre-physical V&V was not always explicit. Table 1 summarizes these

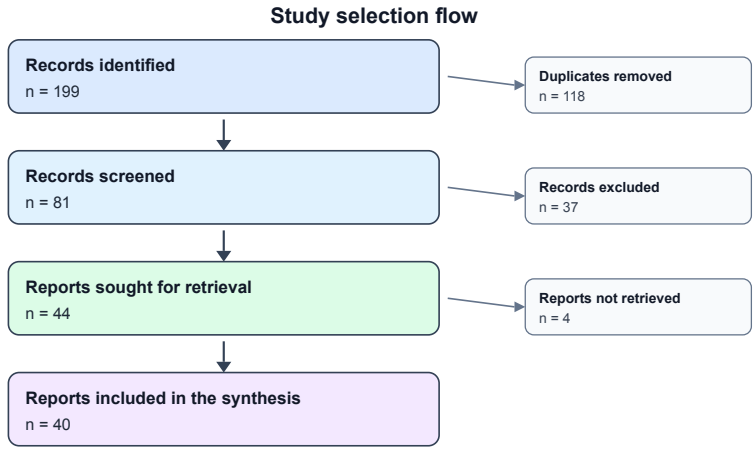


FIGURE 1 Selection flow for the review. The final synthesis contains 40 extracted reports.

report-level descriptors and protocol-specified RQ relevance.

3.2 | RQ1: Modeling generality and EPS representation

EPS relevance was a mutually exclusive report classification. Nineteen reports were coded as *power-budget/electrical-architecture*, 13 as *EPS-adjacent*, 4 as *system-level only*, 2 as *EPS-specific*, and 2 as *not EPS-relevant*. Thus, the largest EPS-relevant category reflects power-budget or electrical-architecture work rather than direct EPS-first modeling.

Model-maturity labels were conceptual framework (18 report memberships), reference model/template (22), prototype demonstration (35), executable model (17), project-applied model (18), and validated/operational model (0). These labels co-occur and cannot support a statistical maturity ranking. Both EPS-specific reports were coded as prototype demonstrations; one also had a reference-model/template label and one an executable-model label, while neither had a project-applied nor a validated/operational label Strobbe et al. (2023); Rehman and Fortin (2024). Figure 2 presents the controlled EPS-relevance and maturity co-reporting.

The reported models were carried principally by block definition/internal block diagrams and parametrics (29 report memberships each), requirements diagrams/tables (28), state/activity/sequence diagrams (23), power budgets and executable simulations (21 each), trade-study models (16), interface-control models (15), and generated documents (12). Test cases/procedures (8), verification matrices (4), and FMEA/risk models (2) were less frequently selected; an additional *other* artifact category was selected in 37 reports. These are overlapping report memberships, not a common minimum artifact set.

Generality was likewise represented through multiple mechanisms: templates (24 report memberships), high-level abstraction (33), method guidance (39), mission-agnostic architecture (22), and low-dynamics/non-detailed-behavior models (10), alongside reusable workflows and other mechanisms. These are reported mechanisms for managing generality or reducing behavioral detail, but the corpus does not demonstrate that minimal prior behavioral knowledge is sufficient to model nanosatellite missions or EPS. Table 2 gives the full controlled memberships.

TABLE 1 Included-report profile, quality, and protocol-specified-RQ relevance.

Measure	Value
Included reports	40
Core / supporting reports	21 / 19
Publication years	2013–2025
Quality score, mean (range)	6.21 (5.0–7.5)
High / medium extraction confidence	29 / 11
Nested claim records	270
RQ1 relevance: high / partial / none	24 / 16 / 0
RQ2 relevance: high / partial / none	3 / 37 / 0
RQ3 relevance: high / partial / none	2 / 29 / 9

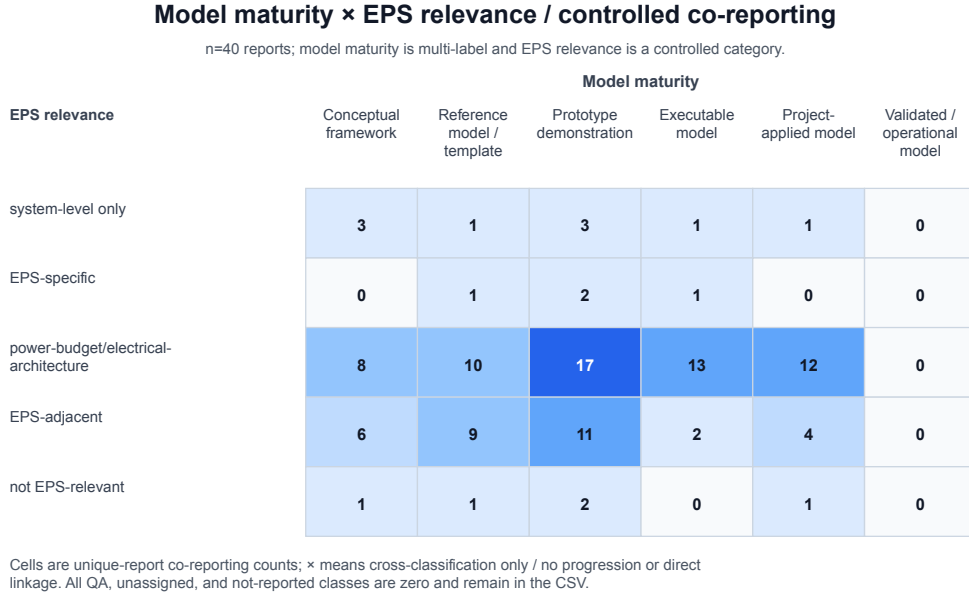


FIGURE 2 Controlled EPS-relevance × model-maturity co-reporting among 40 included reports. Model maturity is multi-label; cells are report memberships and are descriptive cross-classifications only, not an ordinal ranking, progression, or direct linkage.

TABLE 2 Controlled generality and reuse mechanisms. Entries are multi-label report memberships among 40 included reports.

Mechanism	*n*	Mechanism	*n*
Method guidance	39	Reusable workflow	34
High-level abstraction	33	Template	24
Mission-agnostic architecture	22	Reference model	21
Specific-mission model with reusable method only	14	Interface-focused model	12
Library	11	Low-dynamics / non-detailed-behavior model	10
Profile	6	Transformation	6
Open package	6	Pattern	3

The EPS-elements field had status *coded* in 33 of 40 reports; field-status *not reported* (source silence), *not applicable*, and *none* (explicit absence or no use) occurred in 1, 5, and 1 reports, respectively. Among the controlled selections, solar panels and batteries each appeared in 25 report memberships, power budgets in 23, loads in 19, power conditioning/distribution in 18, interfaces in 16, modes and margins in 14 each, degradation in 5, and fault protection in 4. These co-occurring controlled selections describe reported coverage, not complete EPS architectures. Figure 3 displays the selections and their field-status context.

3.3 | Cross-cutting MBSE approach and toolchain continuity

Approach classification is a cross-cutting characterization rather than a fourth research question. The Pure, Enhanced, Extended, and Integrated MBSE/digital-engineering taxonomy was adapted from Buruso et al. (2026). Six reports were classified as Pure MBSE, 11 as Enhanced MBSE, 17 as Extended MBSE, and 6 as Integrated MBSE/digital engineering. This classification describes the reported configuration of modeling and analysis activities; it does not establish comparative effectiveness.

Interoperability mechanisms had field status *coded* in 29 of 40 reports: import/export appeared in 20 report memberships, scripts in 14, spreadsheets in 10, and model transformation in 9. Field-status *not reported* denoted source silence in 10 reports, and 1 report was *uncertain*. Toolchain limitations were coded from author-stated or extractor-identified evidence, rather than necessarily from author-reported limitations alone: missing interfaces appeared in 20 report memberships, scalability issues in 7, licensing/cost in 5, and vendor lock-in and manual translation in 4 each. The limitations field had status *coded* in 25 reports and field-status *not reported* (source silence) in 15. Thus, source silence and a no-selected-controlled-value bar are not evidence of absent mechanisms or limitations. Figure 4 contrasts these controlled memberships.

3.4 | RQ2: Claimed outcomes and evidence

Only 17 of 270 nested claims were EPS-specific. The full 270-claim synthesis is a broader evidence map for nanosatellite MBSE and cannot be read as 270 EPS-derived outcomes. It yielded 736 controlled outcome-domain memberships. The overlapping totals were traceability, consistency, and requirements quality (121); integrability, interoperability, and lifecycle continuity (114); reproducibility, reuse, and openness (87); and reliability, mission assurance, and risk (86). The remaining domains were cost, schedule, effort, efficiency, and automation (85); decision support, analysis,

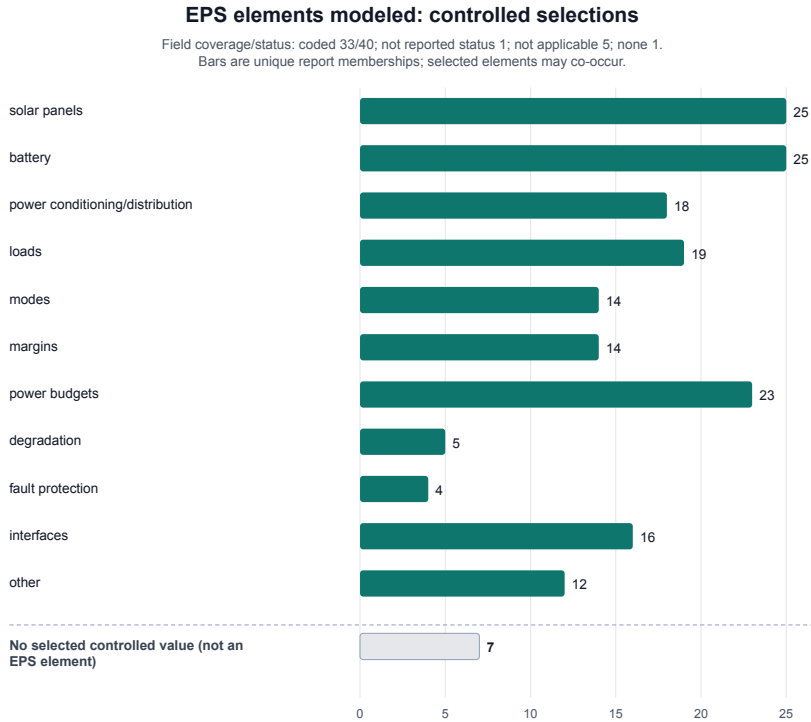


FIGURE 3 Controlled EPS-element selections. Field status was *coded* in 33 of 40 reports; field-status *not reported* (source silence), *not applicable*, and *none* (explicit absence or no use) occurred in 1, 5, and 1 reports, respectively. Bars are report memberships, selections can co-occur, and the neutral no-selected-value bar is not an EPS element.

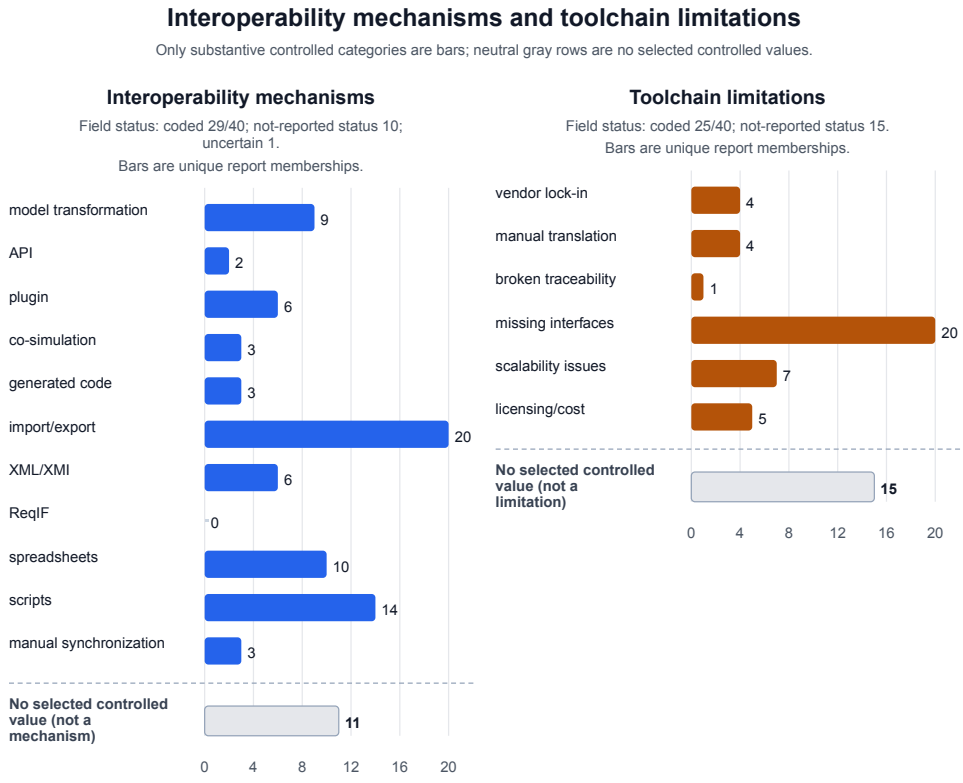


FIGURE 4 Controlled interoperability mechanisms and toolchain limitations. Bars are multi-label report memberships. Interoperability field status was *coded* in 29 of 40 reports, *not reported* (source silence) in 10, and *uncertain* in 1; limitations were *coded* in 25 and *not reported* (source silence) in 15. Neutral bars denote no selected controlled value rather than a substantive category.

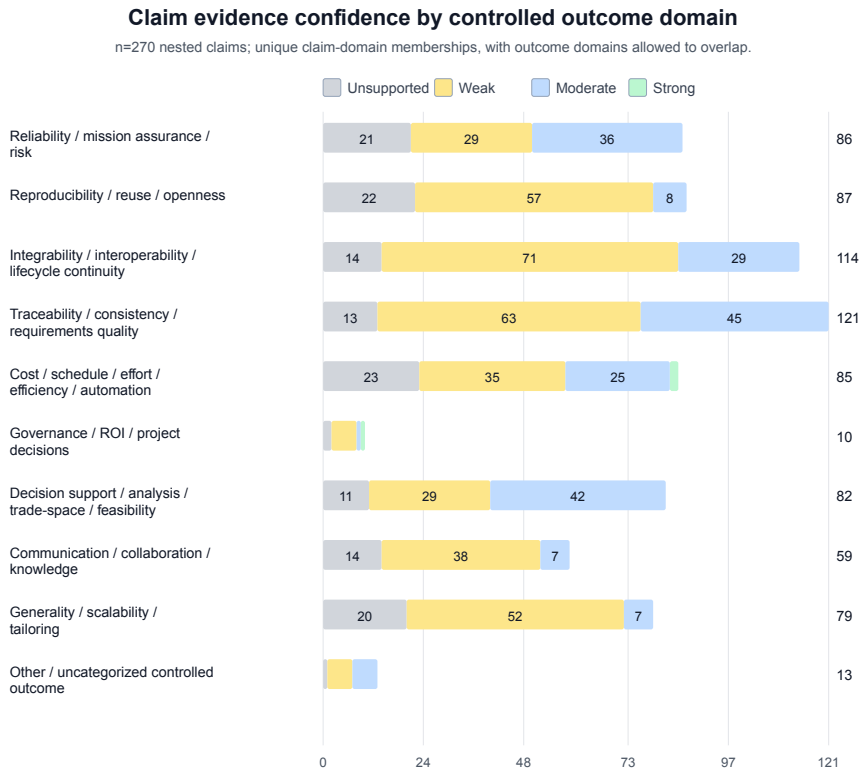


FIGURE 5 Evidence confidence by controlled outcome domain. The figure summarizes 270 nested claims and 736 overlapping claim-domain memberships; confidence segments partition an individual domain row, not the corpus as a whole.

trade-space, and feasibility (82); generality, scalability, and tailoring (79); communication, collaboration, and knowledge (59); governance, ROI, and project decisions (10); and other/uncategorized controlled outcomes (13). These are claim-domain memberships, not report counts.

Evidence confidence was Unsupported for 52 claims, Weak for 139, Moderate for 77, and Strong for 2. The two Strong claims are nested in one report Bella et al. (2024) and therefore do not represent two independent implementations. Figure 5 displays confidence within each outcome domain; domains can overlap across a claim.

Claim specificity was EPS-specific for 17 claims, nanosatellite system-level for 197, generic space-system for 16, generic MBSE for 39, and unclear for 1. Of the 17 EPS-specific claims, 4 were Weak and 13 Moderate; none was Strong. Evidence type is multi-label: illustrative example or conceptual argument was selected for 175 claims, single case-study demonstration for 106, assertion only/no evidence for 72, executable model or simulation result for 36, trade-study or requirements-checking result for 30, quantitative metric without baseline for 18, quantitative comparison with baseline or counterfactual for 3, stakeholder/user/expert evaluation for 6, reuse/replication across projects for 2, and integration/test/operational/flight evidence for 2.

A named comparator category was selected for 53 claims; the selected normalized comparator categories *none* and *not reported* occurred in 35 and 177 claims, respectively. The selected normalized *not reported* category is distinct

from the baseline field-status *not reported* for 22 claims, which denotes source silence; where field-status *none* is used, it denotes explicit absence or no use. Five baseline records contained free-text values outside the controlled vocabulary and were retained as *unmapped/invalid controlled value*, not as author-reported statuses. A stated comparator is not necessarily a quantitative baseline. Figure 6 visualizes the claim-evidence and comparator distinctions.

Reliability subtype results are also claim-level, overlapping memberships rather than report counts or a partition: model consistency/reliability (60), mission assurance (57), defect/error reduction (17), system/mission reliability (11), process reliability (10), EPS reliability (2), and generic reliability claim (2). The not-applicable subtype was selected for 146 claims. The small EPS-reliability count should therefore not be inflated by broader claims about consistency, mission assurance, or process reliability.

3.5 | RQ3: V&V integration before physical development

All RQ3 denominators below are over the 40 included reports and include the nine reports coded RQ3-none. Absence counts are corpus-level context, not prevalence among only RQ3-relevant reports.

The direct pre-physical verification-mechanism field had at least one coded mechanism in 26 reports. Its overlapping memberships were parametric checks (19), requirements-in-the-loop (17), simulation-in-the-loop (15), trade-study verification (15), virtual integration (10), and model checking (0). Twenty-five of the 26 reports co-occurred with an EPS-relevant report classification; this is co-reporting only, not evidence that the mechanism was EPS-specific.

A separate direct V&V-artifact field was coded in 24 reports: simulation scenarios (15), executable checks (14), acceptance criteria (10), test procedures (6), test cases (5), verification matrix (4), SIL/HIL links (2), automated tests (1), and generated test artifacts (0). The separate integration/test-readiness field was coded in 10 reports: software integration (8), links to hardware integration (6), operational validation (4), flat-sat/testbed (0), and environmental test (0). These are overlapping report memberships.

Figure 7 presents a separate derived synthesis: a mixed-source union of V&V artifacts, direct pre-physical mechanisms, and integration/test readiness. For the generic/planned fallback only, controlled RQ3 relevance was additionally used when no coded V&V artifact or pre-physical mechanism was selected. Not every derived category is direct pre-physical evidence. The derived categories were requirements-to-model traceability or verification matrix (19), executable simulation or scenario analysis (22), parametric/budget/constraint checks (23), test cases/procedures/V&V activity packages (21), external tool or data-flow coupling for V&V (11), physical/SIL/HIL/integration-test readiness link (10), EPS/power-relevant V&V artifact/mechanism co-reporting (25), no coded V&V artifact or pre-physical mechanism (12), and generic/planned V&V linkage only (3). A readiness link does not establish closure.

A separate single-select field recorded the highest evidenced IVV integration level for each report. Simulation-in-the-loop was selected for 12 reports; none/not reported for 10; executable/model-based checks for 6; trace-linked verification artifacts for 5; model-contained verification artifacts for 2; separate document only for 1; and software/hardware-in-the-loop for 1. Verification matrix adjacent to model and physical integration/test only were each selected for 0 reports. Three reports had no selected IVV level. The selected controlled *none/not reported* IVV level is distinct from both the IVV field-status *not reported* and those three unassigned values.

Lifecycle-stage labels were multi-label report memberships: concept/pre-Phase A (36), Phase A/B (21), implementation (17), operations (17), integration and test (11), PDR (2), and CDR (2). Figure 8 is controlled co-reporting only; it does not trace reports longitudinally through lifecycle stages or demonstrate a direct lifecycle-to-IVV linkage. In particular, an integration/test or readiness label does not establish completed test closure.

Claim evidence anatomy

n=270 nested claims. Left cells are claim counts; right cells are evidence-type memberships (multi-label).
Comparator recorded = previous document-centric process, manual analysis, alternative toolchain, or previous mission.
All-zero QA rows are omitted from the figure but remain zero in the CSV; the invalid baseline column is retained.

Claim specificity × evidence confidence

	Unsupported	Weak	Moderate	Strong	invalid / unmapped	no selected value
EPS-specific	0	4	13	0	0	0
nanosatellite system-level	27	109	59	2	0	0
generic space-system	7	6	3	0	0	0
generic MBSE	17	20	2	0	0	0
unclear	1	0	0	0	0	0

Evidence type × baseline-comparator availability

	comparator recorded	explicitly none	not reported	invalid / unmapped	no selected value
Assertion only / no evidence	13	4	53	2	0
Illustrative example or conceptual argument	32	18	120	5	0
Single case-study demonstration	23	21	59	3	0
Executable model or simulation result	3	11	22	0	0
Trade-study or requirements-checking result	4	10	16	0	0
Quantitative metric without baseline	3	4	11	0	0
Quantitative comparison with baseline or counterfactual	3	0	0	0	0
Stakeholder/user/expert evaluation	1	1	4	0	0
Reuse/replication across projects	0	1	1	0	0
Integration/test/operational/flight evidence	1	0	1	0	0

Counts are controlled claim memberships; all-zero QA rows are omitted here but remain zero in the CSV.

FIGURE 6 Claim evidence anatomy for 270 nested claims. Claim specificity is single-select, whereas evidence types are multi-label memberships. Comparator categories retain named comparators, selected normalized *none* and *not reported*, and unmapped/invalid values as distinct classes. The selected normalized *not reported* category is distinct from field-status *not reported* (source silence); field-status *none* denotes explicit absence or no use.

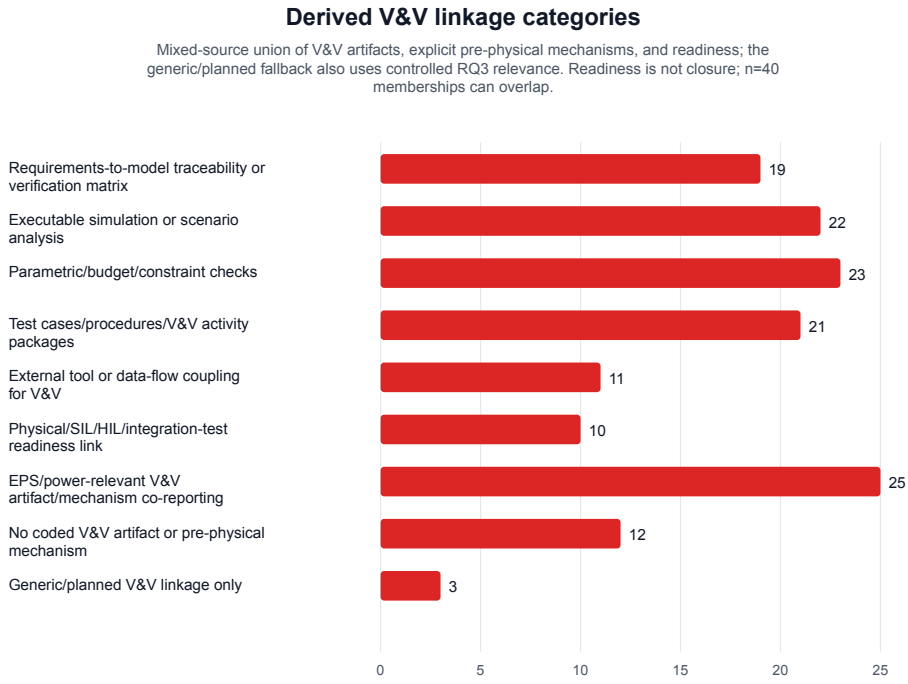


FIGURE 7 Derived V&V linkage categories. Values are overlapping report memberships among 40 included reports. This mixed-source union uses V&V artifacts, direct pre-physical mechanisms, and integration/test readiness; the generic/planned fallback additionally uses controlled RQ3 relevance when no coded artifact or mechanism is selected. Not every category is direct pre-physical evidence, and readiness does not establish closure.

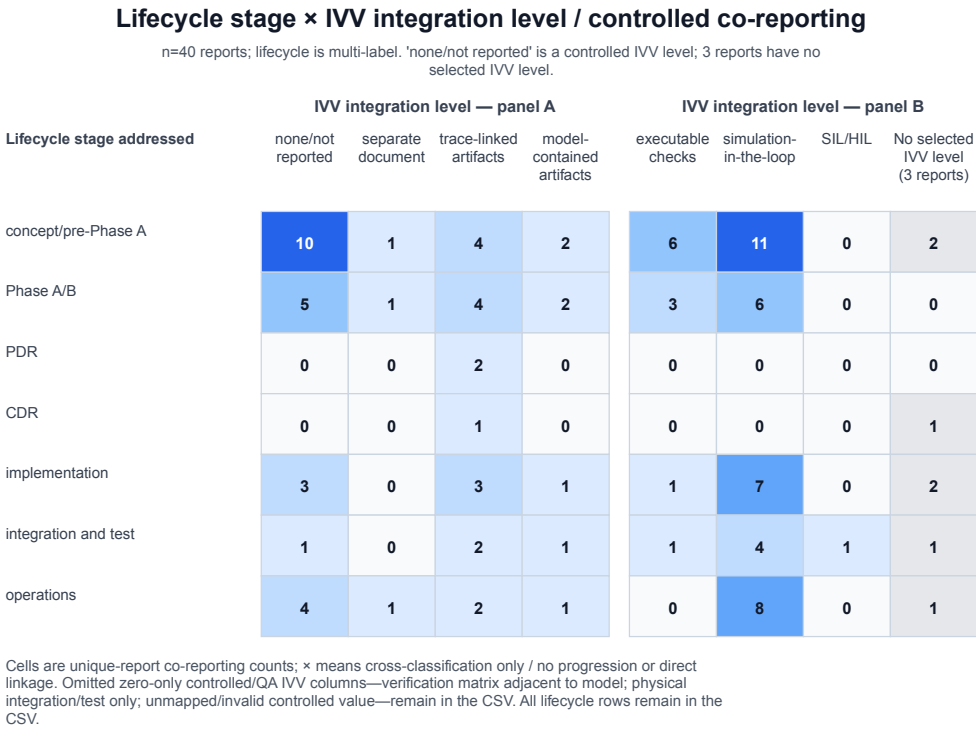


FIGURE 8 Lifecycle-stage × IVV-level controlled co-reporting. Lifecycle labels are multi-label report memberships; cells do not establish longitudinal progression or direct linkage. The selected none/not-reported IVV level is distinct from the three reports with no selected IVV level.

TABLE 3 Buruso-inspired secondary gap mapping. Each row is a mutually exclusive classification across 40 reports; "addressed with evidence" denotes report-level evidence and not proof that a broader gap is resolved.

Gap	Evidence	Conceptual	Mentioned	Not addressed
Practical implementation pathway	29	11	0	0
Toolchain fragmentation/interoperability	19	13	6	2
Lightweight/resource-constrained tailoring	15	21	3	1
Governance/project deliverables/ROI/value	7	20	13	0
NewSpace/agile/distributed/rapid iteration	6	11	15	8

3.6 | Buruso-inspired secondary gap mapping

A secondary mapping adapted the broad space-MBSE gaps identified by Buruso et al. (2026); it did not introduce new research questions. "Addressed with evidence" means addressed with report-level evidence under the controlled classification, not that a field-wide gap has been solved. The mapping is summarized in Table 3.

Direct governance and value-realization fields provide a narrower perspective than the gap mapping. Model-governance strategies were coded in 17 of 40 reports, and ROI/value-realization evidence in 5. These direct fields should not be conflated with the broader governance mapping, which classified 7 reports as addressed with evidence, 20 as addressed conceptually, and 13 as mentioned only.

4 | DISCUSSION

4.1 | Five cautious findings from the evidence chain

System-level and reference-model breadth exceeds direct EPS-specific coverage.

The evidence base includes reusable whole-nanosatellite structures and reference-model work, while only two reports are classified as EPS-specific. The Kaslow/CSRM sequence provides a related publication lineage through reference-model, behavioral, V&V, and system-architecture contributions Kaslow et al. (2014); Spangelo et al. (2013); Anderson et al. (2014); Kaslow et al. (2020); Kaslow and Madni (2017). It is not independent replication and should not be treated as a statistical maturity benchmark. The two EPS-specific reports provide more direct power-subsystem examples, while closely related electrical-architecture work remains in the separate power-budget/electrical-architecture category Strobbe et al. (2023); Boan (2025); Rehman and Fortin (2024). Thus, breadth of system-level representation should not be read as direct evidence of EPS-specific generality or validation. Figure 9 provides a selected evidence map using publication-facing short labels rather than internal identifiers.

The most frequently coded EPS representation is power-budget or electrical architecture.

The largest EPS-relevance category is power-budget/electrical-architecture, and the controlled element selections concentrate on solar generation, battery storage, loads, and power budgets. Interfaces, modes, margins, and power conditioning/distribution also appear, whereas degradation and fault protection are less often coded. These patterns describe this corpus rather than a universal EPS architecture; multi-select element coverage does not show that every report represented each element or an entire EPS.

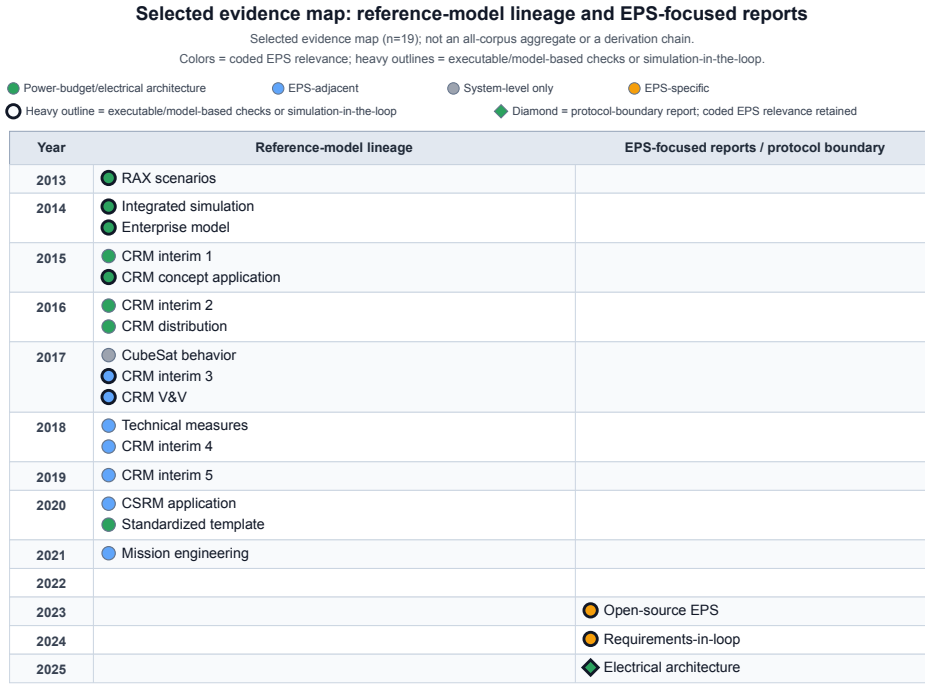


FIGURE 9 Selected evidence map of 19 included reports from the reference-model lineage and EPS-focused/protocol-boundary subset. Publication-facing short labels identify reports; the display is not an all-corpus aggregate, citation-dependency graph, or derivation chain. Colors show coded EPS relevance; outlined symbols show executable/model-based checks or simulation-in-the-loop, and the diamond marks the retained power-budget/electrical-architecture boundary report.

Value claims are common, but baseline-backed or independent outcome evidence is uncommon.
Traceability, integrability, reproducibility, reliability, and decision-support claims are numerous, and executed engineering demonstrations can be useful for design understanding. For example, model-linked simulation and trade studies demonstrate pre-physical power or mission analysis Kaslow et al. (2014); Spangelo et al. (2013); Luccisano et al. (2024); Campioli et al. (2025); Boan (2025); Rehman and Fortin (2024). Such demonstrations should be distinguished from measured project effects. Only three claim memberships used quantitative comparison with a baseline or counterfactual, the two Strong claims were nested in one report Bella et al. (2024), and reuse, integration/test/operational, and stakeholder-evaluation evidence were sparse. Reported comparators therefore warrant cautious interpretation rather than causal claims about cost, schedule, reliability, or integration outcomes.

Pre-physical V&V is present, but reported evidence of continuous model-to-test closure is limited.
Direct mechanism codes and related artifacts indicate reported pre-physical V&V activity. Examples include constraint checks linked to requirements, power-budget analyses, and scenario simulations Strobbe et al. (2023); Hanafi et al. (2024); Rehman and Fortin (2024); Waseem and Sadiq (2018). However, the highest selected IVV levels are usually simulation-in-the-loop, executable checks, or trace-linked artifacts. Readiness does not establish closure, and there is limited reported evidence of model-to-test closure. The results consequently support reported pre-physical analysis, not a general claim of continuous model-to-test verification closure.

Toolchain continuity and governance/value evidence are uneven.
Import/export, scripts, and transformations document attempts to connect modeling and analysis environments, while missing interfaces, scalability, licensing, and translation limitations are also reported. Governance strategies and direct ROI/value-realization evidence are coded in fewer reports than general implementation pathways. These descriptive patterns do not establish a proven bottleneck or identify its cause; they indicate unevenly reported continuity and value evidence in the included corpus.

4.2 | Implications for lightweight nanosatellite MBSE practice

Within these evidence limits, a lightweight approach may be more defensible when it begins with a bounded model thread: mission modes and assumptions, EPS requirements, generation/storage/load and budget abstractions, analysis or executable checks, and the verification artifact needed for the next decision. Reference-model, open-source, and concurrent-design reports illustrate components of such a thread Strobbe et al. (2023); Luccisano et al. (2024); Chiu et al. (2023); Campioli et al. (2025); Buitrago-Leiva et al. (2024). The review does not establish one preferred toolchain or process. Rather, it suggests that teams should make the intended decision, evidence type, ownership, and handoff explicit if they seek evidence beyond an illustrative model.

This framing retains the potential benefit of lightweight tailoring while avoiding an assumption that additional tools create value automatically. A reported readiness link should be treated as preparation for downstream integration or test, not as evidence that verification has been closed. Similarly, governance and value-realization claims require direct evidence rather than inference from the presence of a model or a tool connection.

4.3 | Threats to validity and review limitations

The search was limited to Scopus and IEEE Xplore and to the protocol's publication and language criteria. Relevant grey literature, internal mission evidence, and non-indexed tool repositories may therefore be underrepresented. Four

sought reports were not retrieved, and the synthesis is limited to the 40 included reports.

The controlled coding scheme improves transparency but necessarily simplifies heterogeneous reporting. No independent second assessment or agreement statistic was conducted, and classifications of EPS relevance, IVV integration, outcome evidence, and gap mapping retain reviewer judgment. Status distinctions and the explicit unmapped/invalid class mitigate, but do not remove, ambiguity. Multi-label memberships and descriptive co-reporting also preclude causal or longitudinal inference.

Finally, related publications in the reference-model lineage mean that report counts are not independent implementation counts. Sparse direct EPS-specific coverage, limited baseline-backed value evidence, and limited model-to-test closure constrain generalization beyond the included nanosatellite literature.

5 | CONCLUSION

The included reports describe reusable system-level models, claimed MBSE benefits, and pre-physical analysis and verification activity. The evidence chain thins as the focus narrows to direct EPS scope, baseline-backed value, governance and ROI, and model-to-test closure. These findings describe the available controlled evidence rather than causal effects: direct EPS validation, measured value realization, and continuous verification closure have limited reported evidence.

Data and Supporting Information Availability

Supporting Information contains the protocol, complete search record, controlled codebook, included-report inventory, aggregate tables, and claim-level evidence records used in this review.

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